Research Summary: Secure Multi-Party Sorting and Applications

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Premise

The paper proposes a solution for secure data sorting and aggregation among n mutually distrustful parties $(P_1, \dots, P_n)$. The core setting involves parties who wish to compute a joint function on their private datasets without revealing individual inputs.

- **MP Sorting:** Parties $P_1, \dots, P_n$ hold secret sequences of elements $\boldsymbol{a}^{(1)}, \dots, \boldsymbol{a}^{(n)}$, respectively. The goal is to produce a sorted union of these sequences $\boldsymbol{a}$ without revealing elements of each party to any other party.
- **Weighted Set Intersection:** Inputs are (key, weight) pairs from all parties. The aim is to sum the weights for all inputs with identical keys, and output unique keys with aggregate weights.
- **Top-k & IDS:** The framework is applied to a *Cooperative Intrusion Detection System*. Organizations secretly share logs of (IP address, traffic) to detect distributed attacks (e.g., slow port scans) by identifying heavy hitters (top-k anomalous IPs) without exposing sensitive local traffic data.

Sorting Algorithm

The authors utilize **Batcher's Odd-Even Merge Sort**, a data-oblivious sorting network suitable for MPC environments.

Secure Comparison

- An optimized secure comparision protocol has been followed as described in another research paper. Here, it is assumed to be a blackbox in which shares of two values are inserted as input and random shares of the minimum and maximum elements are given as output.
- **Reference Paper:** *Multiparty Computation for Interval, Equality, and Comparison without Bit-Decomposition Protocol* by *Takashi Nishide, Kazuo Ohta (2007)*

Characteristics & Primitive

- **Data-Oblivious:** The sequence of comparisons is independent of input values, preventing side-channel leakage. Here, the position of the elements is being used rather than the values of the elements themselves.

- **Structure:** It is a sorting network built from fixed **Compare-Exchange** gates. For secret shares $[x]$ and $[y]$, the gate computes $[min]$ and $[max]$ using arithmetic operations on a securely computed comparison bit $[c]$ indicating whether $x < y$.
- **Complexity:** $\mathcal{O}(n \log^2 n)$ comparisons in $\mathcal{O}(\log^2 n)$ communication rounds.

Algorithm 1 Odd-Even Merge

Input: Sequence $\boldsymbol{a}$ whose two halves $a_1, a_2, ...a_{n/2}$ and $a_{n/2+1}, a_{n/2+2}, ...a_n$ are sorted. Length n is a power of 2.
Output: Sequence $\boldsymbol{a}$ is modified in-place to be sorted.

```
if n > 2 then
    Odd-Even Merge(a_1, a_3, a_5, ...a_{n-1})
    Odd-Even Merge(a_2, a_4, a_6, ...a_n)
    for i = 2, 4, 6, ..., n - 2 do
        Compare-Exchange(a_i, a_{i+1})
    end for
else
    Compare-Exchange(a_1, a_2)
end if
```

Algorithm 2 Odd-Even Merge Sort

Input: Sequence $\boldsymbol{a}$ of length n (a power of 2).
Output: Sequence $\boldsymbol{a}$ is modified in-place to be sorted.

```
if n > 1 then
    Odd-Even Merge Sort(a_1, a_2, ..., a_{n/2})
    Odd-Even Merge Sort(a_{n/2+1}, a_{n/2+2}, ..., a_n)
    Odd-Even Merge(a)
end if
```

Protocol & Scheme

- **Framework:** Implementation uses **Sharemind**, a general-purpose MPC platform.
- **Scheme: Additive Secret Sharing** over the ring $\mathbb{Z}_{2^{32}}$.
- **Security Model:** Information-theoretic security against a **passive (honest-but-curious) adversary** corrupting at most one of the three computing parties.

Results

The protocol was tested on a cluster of three commodity servers (Gigabit Ethernet).

- **Performance:** Successfully sorted 2^{14} **elements** in a little more than **3.5 minutes**. The implementation is moderately optimized, implying that this performance has scope for improvement.
- **Conclusion:** While slower than plaintext sorting, the method is feasible for high-latency batch processing tasks like inter-organizational log analysis.